

Saaram: Resource-Aware Telugu News Summarization with Morphology-Aware TF-IDF and mT5

Hariharan Narlakanti¹, Vivek Nidumolu¹, Vishnu Vardhan Reddy Padala¹,
Sanjeev Pratur¹, and Vinay Simha Reddy Tappeta¹

Department of Computer Science and Engineering (AI & ML),
Malla Reddy University, Hyderabad, Telangana, India
narlakantihariharan@gmail.com

Abstract Telugu is a morphologically rich, agglutinative language that poses significant challenges for tokenization and evaluation in standard NLP pipelines. We present *Saaram* (సారం, “essence”), a deployed Telugu news summarization service with optional speech synthesis, and conduct a comparative study of five configurations evaluated on the complete 1,302-sample XL-Sum Telugu test split. Our morphology-aware TF-IDF baseline replaces word-level tokenization with character n -gram analysis (`analyzer='char_wb'`, $n \in [2, 4]$) and lead-bias position weighting, achieving a 26.6% relative improvement in ROUGE-2 over the word-level baseline. Evaluated abstractive models show that the 8-bit LoRA fine-tuned mT5 model achieves the highest lexical overlap (ROUGE-2: 0.0496) but yields near-identical semantic quality, measured by BERTScore (0.7272), compared with mT5 Base (0.7273), indicating that in-distribution fine-tuning provides negligible semantic gains. Finally, a resource-aware runtime router adaptively dispatches requests between TF-IDF and mT5 Base, retaining 99.56% of mT5 semantic quality (BERTScore) while mitigating memory exhaustion risk. Latency benchmarking further characterizes deployment-relevant cost–quality trade-offs under resource constraints.

Keywords: Telugu text summarization · morphology-aware TF-IDF · mT5 · low-resource NLP · resource-aware routing

1 Introduction

Telugu, with over 80 million native speakers across Telangana and Andhra Pradesh, is one of the most widely spoken languages in India, yet remains severely underrepresented in deployed NLP systems [1]. The rapid proliferation of digital Telugu news creates a clear societal need: readers face an overwhelming volume of content, while few publicly accessible automated summarization systems exist for Telugu at scale.

Two structural challenges compound this problem. First, Telugu is an *agglutinative* language, where a single root word can generate dozens of inflected surface forms through affixation. Standard NLP pipelines that assume word-level tokenization—primarily designed for morphologically simpler languages such as English—treat these inflected forms as unrelated vocabulary items, severely underestimating term importance and fragmenting evaluation metrics [2]. For example, the root రాష్ట్రం (state) appears in news text as రాష్ట్రంలో (in the state) and రాష్ట్రానికి (to the state), yet word-level TF-IDF treats all three as entirely unrelated tokens. Second, many standard ROUGE implementations strip non-ASCII characters by default, rendering Telugu ROUGE evaluation invalid without explicit adaptation.

Prior Telugu NLP research has addressed these challenges in isolation: extractive approaches without morphological awareness [5], or neural abstractive models without deployment-oriented analysis [4]. To the best of our knowledge, no prior Telugu summarization study jointly evaluates morphology-aware extractive methods, large-scale neural summarization, and production deployment trade-offs within a unified framework.

This paper presents *Saaram* (సారం, “essence”), a deployed Telugu news summarization system with optional speech synthesis, and makes five contributions:

- A **morphology-aware TF-IDF baseline** (TF-IDF V2) using character n -grams and lead-bias weighting, achieving a 26.6% relative improvement in ROUGE-2 over word-level TF-IDF V1.
- A **large-scale comparative benchmark** evaluating five summarization configurations on the complete 1,302-sample XL-Sum Telugu test split using ROUGE, BERTScore, and latency profiling.
- A **LoRA fine-tuning ablation study** showing that 8-bit quantized LoRA adaptation yields only marginal lexical gains (+0.0032 ROUGE-L) without semantic improvement (BERTScore: 0.7272 vs. 0.7273) over the pre-trained base model.
- A **resource-aware inference strategy** that routes requests at runtime while retaining 99.56% of mT5 semantic quality (BERTScore), routing only 5.38% of samples (70 of 1,302) to TF-IDF V2 as a robust fallback under resource constraints.
- A **resource-constrained deployment analysis** on free-tier CPU cloud infrastructure, characterizing memory-aware lazy loading, fallback behavior, and TTS latency.

The complete system is publicly deployed, with the frontend hosted on Vercel, the backend on Hugging Face Spaces (Docker, CPU-only), and the source code available on GitHub.

2 Related Work

Extractive and neural abstractive summarization. Graph-based methods such as LexRank [6] and TextRank [7] rank sentence salience using centrality

measures, while MMR [8] introduces diversity-aware reranking. TF-IDF-based sentence scoring remains a classical baseline for both English and Indic summarization tasks [1]. Prior Telugu extractive work [5] primarily relies on word-level tokenization, which does not adequately capture agglutinative morphology. In abstractive summarization, attention-based architectures [9], pointer-copy mechanisms [10], and coverage penalties [11] have improved sequence-to-sequence generation. The Transformer [12] and T5 [13] further enabled unified text-to-text modeling. Multilingual T5 (mT5) [14] extends this framework to 101 languages, using SentencePiece-based subword segmentation that better accommodates Telugu morphology.

Indic and Telugu NLP. IndicBART [15] is a pretrained model for Indic text generation, although our preliminary qualitative screening revealed instability on Telugu summarization inputs (Sect. 3.4). Doddapaneni et al. [16] survey Indic language models and identify data scarcity as a primary bottleneck, which datasets such as XL-Sum [3] and TeSum [19] help alleviate. Prior Telugu summarization work [5] applied TextRank and MMR while retaining word-level tokenization, a limitation addressed in this study.

Evaluation for agglutinative languages. ROUGE [17] measures exact n -gram overlap, often penalizing inflected paraphrases common in agglutinative languages. BERTScore [18] uses pretrained contextual embeddings to compute token similarity, theoretically assigning partial credit to semantically equivalent yet morphologically varied tokens. Motivated by this property, we use BERTScore as the primary semantic similarity metric while reporting ROUGE as a supplementary lexical overlap measure.

Deployment gap. Indic NLP literature rarely reports real-world deployment characteristics. Existing systems typically emphasize quality metrics without analyzing runtime latency, memory constraints, or resource-aware fallback mechanisms. This gap is addressed in our work through comprehensive deployment-aware profiling.

3 System Architecture

Saaram implements a three-tier architecture designed under two practical deployment constraints: free-tier cloud hosting with limited memory and production CPU-only serving constraints, together with interactive latency requirements for user-facing summarization. Fig. 1 illustrates the complete pipeline. The frontend exposes three inference quality tiers corresponding to TF-IDF V2, mT5 Base, and mT5 Fine-tuned models. Optional Edge TTS audio synthesis is supported for spoken summaries.

3.1 Input Processing

The system supports three input modes: (1) direct text entry, (2) URL-based article extraction using BeautifulSoup4 with a 10-second timeout and `<p>`-element

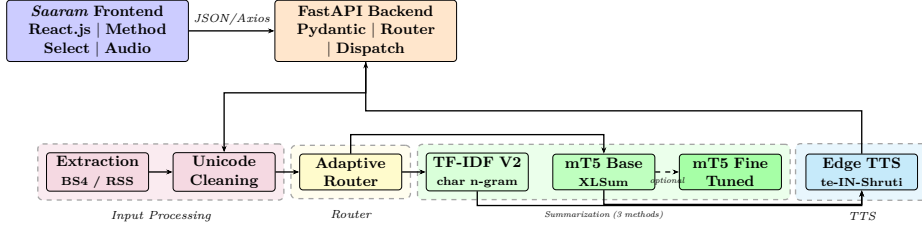


Figure 1. *Saaram* three-tier architecture. The resource-aware router selects TF-IDF V2 or mT5 Base at runtime; mT5 Fine-tuned (8-bit LoRA) is available as a manual override. Edge TTS converts summaries to Telugu audio asynchronously.

prioritization, and (3) live RSS ingestion from BBC Telugu for the Saaram Radio bulletin feature. All inputs pass through a Unicode-aware regular-expression cleaning pipeline that preserves the Telugu Unicode block (U+0C00–U+0C7F), dandas (U+0964), ASCII digits, and standard punctuation. This preprocessing pipeline is applied identically during fine-tuning and inference to ensure training–inference consistency.

3.2 Morphology-Aware TF-IDF (V2)

Standard word-level TF-IDF (V1) fails for Telugu because agglutinative inflection creates distinct surface tokens for semantically related forms. Consider the root రాష్ట్రం (state): its inflected forms రాష్ట్రంలో (in the state) and రాష్ట్రానికి (to the state) share no whitespace- delimited word token, causing TF-IDF to assign separate, low term weights to each form of a high-frequency concept.

TF-IDF V2 addresses this with four targeted improvements:

(1) **Character n-gram analysis:** replacing word tokenization with character n-grams (`analyzer='char_wb'`, `ngram_range = (2,4)`) to capture root substrings across inflected forms, restoring term-frequency coherence; (2) **Lead-bias position weighting:** applying decaying position weights to scores to boost earlier sentences, reflecting news inverted-pyramid structures; (3) **Adaptive sentence selection:** dynamically adjusting the sentence extraction count k based on input length to prevent under/over-extraction; and (4) **Per-sentence score exposure:** exposing sentence-level scores to the router (Sect. 3.5) as confidence signals.

3.3 mT5 Abstractive Summarization

mT5 Base. We use the `csebuetnlp/mT5_multilingual_XLSum` checkpoint (580M parameters), pretrained on multilingual mC4 and further adapted for multilingual summarization on the XL-Sum benchmark, including Telugu. SentencePiece tokenization naturally captures Telugu subword morphology. Inference uses beam search with width 4, length penalty 2.0, trigram blocking, output length bounds of 30–128 tokens, and a maximum input length of 512 tokens.

mT5 Fine-tuned (8-bit LoRA). The base checkpoint is further adapted to XL-Sum Telugu using parameter-efficient LoRA [20] applied to an 8-bit quantized model [21]. LoRA uses rank $r = 16$, scaling factor $\alpha = 32$, target modules $[q, v]$, and dropout 0.05.

Training was performed for 3 epochs with learning rate 3×10^{-4} and per-device batch size 4 with gradient accumulation $\times 2$ (effective batch size 8) on a single NVIDIA Tesla T4 GPU (16 GB VRAM, Kaggle environment). Training input and target lengths were capped at 384 and 96 tokens, respectively, while deployed inference uses `max_length=128`. The dataset was split into 10,421 training, 1,302 validation, and 1,302 test samples. An earlier full-parameter fine-tuned baseline (3.372s latency) is reported in Table 2 for reference.

Since the base checkpoint already contains Telugu summarization knowledge from XL-Sum pre-adaptation, additional LoRA fine-tuning yields only marginal lexical improvements (ROUGE-L +0.0032) without measurable semantic gains (BERTScore: 0.7272 vs. 0.7273).

3.4 Model Backbone Selection

Four abstractive backbones were screened qualitatively using a representative set of Telugu news articles, evaluated by two native Telugu speakers and one NLP researcher.

IndicBART (244M parameters) produced unstable outputs, frequently exhibiting token repetition, language switching, and named-entity hallucinations. **mBART-50** (611M parameters) generated grammatically valid summaries but often produced generic or less informative outputs. **mT5 Base** (580M parameters) produced fluent, language-consistent summaries with strong semantic preservation. **mT5 Fine-tuned** (580M parameters) showed similar fluency while providing slightly improved domain consistency.

Based on this qualitative screening, the two mT5 variants were selected for full quantitative evaluation.

3.5 Resource-Aware Inference Strategy

As a deployment optimization layer, the system includes a lightweight inference router that intercepts each summarization request before model dispatch and selects between TF-IDF V2 and mT5 Base without user intervention.

Routing decisions are based on three signals: (1) **Word count**, where inputs below 75 words are routed to TF-IDF V2 and inputs above 200 words to mT5 Base; (2) **TF-IDF score variance**, where variance above 0.04 favors TF-IDF V2 and variance below 0.008 favors mT5 Base; and (3) **Available RAM**, where low-memory conditions force TF-IDF V2 regardless of other signals.

The router never selects the fine-tuned model automatically; fine-tuned inference remains a manual override reserved for cases where marginal ROUGE gains are desired.

On the XL-Sum test split, 70 of 1,302 samples (5.38%) were routed to TF-IDF V2, reflecting the dataset bias toward longer news articles. Since the router

activates for only a small fraction of in-distribution samples, it primarily serves as a deployment safeguard for graceful degradation under resource constraints rather than as a core algorithmic contribution.

3.6 Deployment Constraints and Optimization

Saaram is deployed on Hugging Face Spaces under free-tier constraints (2 vCPU, 16 GB RAM, CPU-only inference), imposing strict resource limitations that required explicit architectural optimizations.

Memory-aware lazy loading. The mT5 checkpoints require approximately 2.3 GB RAM for the 8-bit LoRA checkpoint and 3.7 GB for the earlier full-parameter baseline. Loading multiple checkpoints simultaneously alongside FastAPI, the TF-IDF pipeline, and operating-system overhead risks out-of-memory (OOM) failures. To mitigate this, models are initialized only upon first request (*lazy loading*) and garbage- collected after a configurable idle timeout. As a result, only the actively requested model occupies memory at any given time.

Robust fallback hierarchy. The system handles three major failure modes through automatic degradation. First, OOM events during mT5 inference trigger immediate fallback to TF-IDF V2, with the API returning a flag indicating degraded mode. Second, checkpoint loading failures trigger fallback from the fine-tuned model to mT5 Base, and subsequently to TF-IDF V2 if required. Third, URL extraction failures return structured error responses with clear recovery guidance. These safeguards ensure that the system never returns an unhandled empty response.

API design. The backend exposes seven FastAPI endpoints across Health, News, Summarization, and Audio services. Request schemas use Pydantic validation, and invalid requests return structured HTTP 422 error responses for improved debuggability and client-side handling.

4 Experimental Setup

Dataset. All experiments use the complete XL-Sum Telugu test split [3], consisting of 1,302 BBC Telugu news articles paired with human-written reference summaries across politics, business, sports, and current affairs. The base mT5 checkpoint was pretrained on multilingual mC4 and subsequently fine-tuned on the multilingual XL-Sum corpus, including Telugu, making this evaluation largely in-distribution (discussed in Sect. 6).

Evaluation pipeline. The evaluation pipeline processes the full 1,302-sample test split with checkpointing/resume support, structured CSV outputs, and mean/P95/P99 latency profiling.

ROUGE adaptation. Standard ROUGE scorers strip non-ASCII characters, making Telugu evaluation invalid. We therefore implement a custom Unicode-preserving tokenizer for reliable ROUGE computation over Telugu text.

Preprocessing consistency. All configurations receive identical cleaned inputs using the preprocessing pipeline described in Sect. 3.1. The same preprocessing settings are applied during evaluation and baseline scoring.

Metrics. We report ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore F1. BERTScore is computed using `bert-base-multilingual-cased` with default rescaling for Telugu. Wall-clock latency, measured from request ingestion to output serialization, is averaged across the test split.

5 Results

5.1 TF-IDF V1 vs V2: Morphological Improvement

We isolate the contribution of morphology-aware character n -gram analysis by comparing TF-IDF V1 (word-level) and TF-IDF V2 (character n -gram) in Tables 1 and 2. TF-IDF V2 consistently improves all ROUGE metrics while preserving near-zero inference cost.

ROUGE-2 shows the largest relative gain, improving by 26.6% ($0.0128 \rightarrow 0.0162$), suggesting that character n -grams mitigate bigram fragmentation caused by Telugu morphological inflection. ROUGE-1 and ROUGE-L improve by 14.4% ($0.0709 \rightarrow 0.0811$) and 13.4% ($0.0554 \rightarrow 0.0628$), respectively. Despite these gains, mean latency remains effectively negligible at 0.005 s.

5.2 Full Comparative Evaluation

Table 1 summarizes quality metrics across all five evaluated configurations, enabling direct comparison of extractive, abstractive, and hybrid approaches. Table 2 reports inference latency separately to preserve focus on summarization quality in the primary comparison.

Table 1. Quality Evaluation — XL-Sum Telugu Test Split (1,302 samples). Bold indicates best per metric.

Method	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
TF-IDF V1 (word-level)	0.0709	0.0128	0.0554	0.6626
TF-IDF V2 (char n -gram)	0.0811	0.0162	0.0628	0.6671
mT5 Base (pretrained)	0.1610	0.0478	0.1420	0.7273
mT5 Fine-tuned [†]	0.1639	0.0496	0.1452	0.7272
Auto Router (hybrid)	0.1562	0.0458	0.1372	0.7241

ROUGE values are reported as F1 scores. BERTScore F1 uses `bert-base-multilingual-cased` with rescaling. [†] Fine-tuned model uses 8-bit LoRA adaptation.

(F1) mT5 significantly outperforms TF-IDF. mT5 Base achieves ROUGE-L of 0.1420 compared with 0.0628 for TF-IDF V2, representing a 126% relative improvement and aligning with prior multilingual summarization baselines [3]. BERTScore similarly improves from 0.6671 to 0.7273. TF-IDF V2 consistently outperforms TF-IDF V1, confirming that morphology-aware tokenization improves extractive summarization quality.

Table 2. Inference Latency Profile — Local MPS-accelerated evaluation (1,302 samples).

Method	Mean Latency (s)
TF-IDF V1	0.001
TF-IDF V2	0.005
mT5 Base	2.583
mT5 Fine-tuned [†]	2.482
Auto Router	3.130

[†] Fine-tuned model uses 8-bit LoRA adaptation. Latency reduced versus the earlier full-parameter fine-tuned checkpoint (3.372s; Sect. 3.3).

(F2) LoRA fine-tuning provides negligible semantic gain. mT5 Fine-tuned (8-bit LoRA) yields near-identical semantic quality to mT5 Base (BERTScore: 0.7272 vs. 0.7273), indicating that additional adaptation on an already covered distribution primarily affects lexical realization rather than semantic representation.

(F3) Inference latency remains practical. The 8-bit LoRA fine-tuned model achieves mean latency of 2.482s, slightly faster than mT5 Base (2.583s) and substantially faster than the earlier full-parameter fine-tuned baseline (3.372s).

(F4) Resource-aware routing retains 99.56% of mT5 quality. The router preserves 99.56% of mT5 quality (BERTScore: 0.7241 vs. 0.7273) while routing only 5.38% of samples to TF-IDF V2, serving as an effective fallback mechanism under resource-constrained deployment.

6 Discussion and Analysis

In-Distribution Fine-Tuning. The near-identical performance of mT5 Base and mT5 Fine-tuned (8-bit LoRA) indicates that local LoRA adaptation yields negligible semantic improvement over the base model. Since the base checkpoint (csebuetnlp/mT5_multilingual_XLSum) was already fine-tuned on the multilingual XL-Sum corpus, including Telugu, further in-distribution adaptation primarily alters surface phrasing rather than underlying semantic representations. Practitioners may therefore benefit more from fine-tuning on out-of-distribution domains where pretrained representations are weaker.

Morphology-Aware Character N-Grams. The 26.6% ROUGE-2 improvement of TF-IDF V2 over TF-IDF V1 highlights the value of morphology-aware tokenization. In Telugu, a concept such as *government* can appear as inflected surface forms (ప్రభుత్వం, ప్రభుత్వానికి, ప్రభుత్వంపై) that share the root sequence ప్రభుత్వ. Character n -grams with range (2, 4) capture these shared substrings, aggregating term-frequency signals that word-level TF-IDF would otherwise fragment across distinct vocabulary entries, thereby improving salience estimation.

Router Optimization. The inference router functions primarily as a deployment safeguard, routing only 5.38% of XL-Sum test samples (70 of 1,302) to TF-IDF V2. Because most in-distribution news articles are sufficiently long and

information-dense, mT5 Base remains the dominant inference path. The router therefore serves mainly as an engineering fallback that improves robustness by enabling graceful degradation from neural to statistical summarization under resource constraints.

BERTScore Suitability. BERTScore computes contextual token similarity and can assign partial credit to paraphrases that preserve meaning despite lexical variation. This property is especially relevant for morphologically agglutinative languages such as Telugu, where the same concept may appear in multiple inflected forms. We therefore use BERTScore as the primary semantic similarity metric, while acknowledging that its practical suitability relative to ROUGE for Telugu remains unverified without human evaluation. Such validation is an important direction for future work.

7 Qualitative Analysis

Table 3 presents a qualitative comparison on a representative sample from the XL-Sum Telugu test split (sample ID: `international-53333964`). System outputs were generated using the local *Saaram* evaluation pipeline under the same preprocessing and inference settings used for quantitative evaluation. The table shows top-ranked extractive sentences for TF-IDF variants and decoded outputs for mT5 variants, illustrating the qualitative difference between extraction and abstraction.

Table 3. Qualitative Output Comparison — XL-Sum Telugu test sample (index 0, `international-53333964`). Outputs are actual system inference results.

Source	Telugu Script
Input article	ఎఫ్బీఐ డైరెక్టర్ క్రిస్టఫర్ రే “చైనా ప్రభుత్వ గూఢచర్యం, డేటా చోరీ వల్ల అమెరికా భవిష్యత్తుకు ఎప్పుడూ లేనంత దీర్ఘకాలిక ముప్పు ఉంది” అని అన్నారు. “చైనా చాలా స్థాయిల్లో ఆపరేషన్లు నిర్వహిస్తోంది.” <i>[Article continues; full text available in XL-Sum Telugu dataset, sample ID: international-53333964.]</i>
Gold reference	చైనా ప్రభుత్వం నుంచి అమెరికాకు అతిపెద్ద దీర్ఘకాలిక ముప్పు పొంచి ఉందని అమెరికా నిఘా ఏజెన్సీ ఎఫ్బీఐ డైరెక్టర్ క్రిస్టఫర్ రే అన్నారు.
TF-IDF V1	ఎఫ్బీఐ డైరెక్టర్ క్రిస్టఫర్ రే “చైనా ప్రభుత్వ గూఢచర్యం, డేటా చోరీ వల్ల అమెరికా భవిష్యత్తుకు ఎప్పుడూ లేనంత దీర్ఘకాలిక ముప్పు ఉంది” అని అన్నారు... <i>[Output truncated for brevity]</i>
TF-IDF V2	ఎఫ్బీఐ డైరెక్టర్ క్రిస్టఫర్ రే “చైనా ప్రభుత్వ గూఢచర్యం, డేటా చోరీ వల్ల అమెరికా భవిష్యత్తుకు ఎప్పుడూ లేనంత దీర్ఘకాలిక ముప్పు ఉంది” అని అన్నారు... <i>[Output truncated for brevity]</i>
mT5 Base	చైనా ఆర్థిక గూఢచర్యం, డేటా చోరీ వల్ల అమెరికా భవిష్యత్తుకు ఎప్పుడూ లేనంత దీర్ఘకాలిక ముప్పు ఉందని ఫైనాన్షియల్ బ్యాంక్ ఆఫ్ ఇండియా (ఎఫ్బీఐ) డైరెక్టర్ చెప్పారు.
mT5 Fine-tuned (8-bit LoRA)	చైనా ప్రభుత్వం గూఢచర్యం, డేటా చోరీ వల్ల అమెరికా భవిష్యత్తుకు ఎప్పుడూ లేనంత దీర్ఘకాలిక ముప్పు ఉందని ఎఫ్బీఐ డైరెక్టర్ తెలిపారు.

Three qualitative patterns are visible. First, TF-IDF V1 and TF-IDF V2 both return a near-verbatim first sentence of the article, reflecting the lead-bias

position weighting in V2 and the high term frequency of salient government-related tokens in both methods. Their stronger ROUGE-2 performance in aggregate evaluation (Table 1) therefore arises largely from lexical bigram copying rather than true abstractive compression.

Second, both mT5 variants produce abstractive paraphrases that deviate lexically from the source and reference, helping explain their lower ROUGE-2 despite higher BERTScore. Notably, the mT5 Base output expands the abbreviation *FBI* as ఫైనాన్షియల్ బ్యాంక్ ఆఫ్ ఇండియా (Financial Bank of India), a named-entity hallucination absent from the source. This illustrates a common hallucination failure mode in multilingual abstractive summarization for abbreviation-dense news text.

Third, the mT5 Fine-tuned (8-bit LoRA) output more closely mirrors the noun-phrase structure of the reference summary, consistent with the marginal ROUGE-L gain (+0.0032) without meaningful BERTScore improvement. This suggests that local fine-tuning primarily refines surface phrasing rather than semantic content.

8 Limitations and Future Work

This study has several limitations. (1) **No human evaluation:** all evaluations rely on automatic metrics without human assessment of fluency, faithfulness, or factual consistency; (2) **No component ablation:** the individual contributions of character n -gram analysis, position weighting, and adaptive sentence selection in TF-IDF V2 were not isolated; (3) **TTS quality:** speech synthesis quality was assessed only qualitatively, without a controlled Mean Opinion Score (MOS) study; (4) **In-distribution fine-tuning:** LoRA adaptation was restricted to the news domain, where base model representations were already strong; and (5) **Domain restriction:** generalizability to other Telugu text genres remains unverified.

Future work includes out-of-distribution LoRA fine-tuning (e.g., legal, educational, or conversational Telugu), component-wise ablation of TF-IDF V2, longer-context architectures for documents exceeding 512 tokens, human correlation studies for metric validation, and extension to other Dravidian languages.

9 Conclusion

This paper presented *Saaram*, a deployed Telugu news summarization system with optional speech synthesis, along with a comparative study on the complete 1,302-sample XL-Sum Telugu test split. Our findings show that: (1) morphology-aware TF-IDF V2 yields a 26.6% ROUGE-2 improvement over TF-IDF V1, validating character n -grams for agglutinative languages; (2) in-distribution 8-bit LoRA fine-tuning provides negligible semantic gains over the pre-trained mT5 base model; (3) BERTScore provides a more semantically informative evaluation signal than ROUGE for inflected Telugu outputs, although human validation remains necessary; and (4) the resource-aware router preserves 99.56% of mT5

quality while providing a reliable fallback under memory-constrained deployment settings.

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